📘 Lecture 6 – Processes (25 MCQs)

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1. A thread is best described as:

A) A hardware processor
B) A lightweight software-based processor
C) A full operating system
D) A memory protection unit

Answer: B

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2. A process is defined as:

A) A single thread
B) A software container in which one or more threads run
C) A hardware abstraction
D) A scheduling algorithm

Answer: B

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3. Processor context includes:

A) Only application data
B) CPU register values such as the program counter
C) Thread scheduling policies
D) Memory allocation tables

Answer: B

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4. Thread context consists of:

A) Only user-level data
B) Processor context plus thread-specific information
C) MMU settings only
D) Kernel data structures only

Answer: B

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5. Process context differs from thread context because it includes:

A) User interface state
B) Cache memory
C) Memory management unit (MMU) settings
D) Network buffers

Answer: C

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6. Switching between threads is generally cheaper than switching between processes because:

A) Threads do not share memory
B) Process switching avoids kernel mode
C) Thread switching can be done without OS involvement
D) Threads require more hardware support

Answer: C

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7. Creating and destroying threads is:

A) More expensive than processes
B) Cheaper than processes
C) Equal in cost to processes
D) Impossible without kernel support

Answer: B

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8. One reason to use threads is to:

A) Increase process isolation
B) Avoid blocking during I/O
C) Eliminate context switching
D) Replace the operating system

Answer: B

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9. User mode is characterized by:

A) Full access to hardware
B) Execution of operating system code
C) Restricted access to system resources
D) Ability to manage memory directly

Answer: C

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10. Kernel mode is required when:

A) Running application code
B) Executing system calls
C) Performing user-level scheduling
D) Rendering user interfaces

Answer: B

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11. A major trade-off when using threads instead of processes is:

A) Slower communication
B) Reduced parallelism
C) Higher risk of data corruption
D) Increased isolation

Answer: C

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12. Which is an indirect cost of context switching?

A) Saving CPU registers
B) Executing the interrupt handler
C) Cache disruption
D) Switching stack pointers

Answer: C

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13. User-level threads are efficient because:

A) They always use kernel scheduling
B) All operations are handled inside a single process
C) They eliminate blocking
D) They isolate memory

Answer: B

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14. A major drawback of user-level threads is that:

A) They cannot run in parallel
B) Blocking a single thread may block the entire process
C) They require kernel traps for every operation
D) They cannot be scheduled

Answer: B

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15. Kernel-level threads solve blocking problems because:

A) The kernel schedules processes only
B) The kernel can schedule other threads in the same process
C) They avoid system calls
D) They do not use shared memory

Answer: B

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16. The main disadvantage of kernel-level threads is:

A) Lack of parallelism
B) Increased complexity in applications
C) Reduced efficiency due to kernel traps
D) Inability to handle external events

Answer: C

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17. The two-level threading model introduces:

A) Only user-level threads
B) Only kernel-level threads
C) Kernel threads that execute user-level threads
D) Threads without OS support

Answer: C

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18. In the two-level model, user threads are scheduled by:

A) The kernel scheduler only
B) Hardware interrupts
C) A user-level scheduler within a kernel thread
D) The network layer

Answer: C

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19. Multithreaded web clients mainly use threads to:

A) Improve security
B) Achieve full CPU parallelism
C) Hide network latency
D) Reduce memory usage

Answer: C

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20. Thread-Level Parallelism (TLP) measures:

A) Number of CPUs
B) Degree of instruction-level parallelism
C) Average number of concurrently executing threads
D) Network throughput

Answer: C

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21. A typical web browser’s TLP value indicates that threads are mostly used for:

A) Maximum parallel execution
B) Load balancing
C) Logical organization of tasks
D) Memory isolation

Answer: C

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22. On the server side, multithreading improves performance because:

A) Threads eliminate I/O
B) Starting threads is cheaper than starting processes
C) Servers become stateless
D) Context switching is eliminated

Answer: B

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23. The dispatcher/worker model uses:

A) One thread for all requests
B) Nonblocking system calls only
C) A dispatcher thread assigning work to worker threads
D) Separate processes for each request

Answer: C

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24. Virtualization helps distributed systems mainly by:

A) Reducing network latency
B) Eliminating hardware changes
C) Improving portability and isolation
D) Increasing instruction speed

Answer: C

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25. Which virtualization approach runs directly on hardware with high performance?

A) Process VM
B) Hosted VMM
C) Native VMM
D) Application VM

Answer: C